

All collected data are first transcribed using a competitive ASR system (Gao et al. 2022)⁴, then the ASR 1-bests are processed with punctuation insertion, paragraph segmentation (Zhang et al. 2021a), and speaker diarization (Zheng et al. 2023). This ASR structure is widely utilized and operates at peak performance while preserving the overall generality of the dataset. We utilize GPT-4 (gpt-4-0125-preview)⁵ to obtain the initial target of CoS2W for the SWAB dataset. The input is chunk-level to leverage local context and to reduce token consumption. The auxiliary information is also provided to enhance the ASR error correction performance. The prompt for this procedure is in the appendix.

The results of GPT-4 still contain some issues (as shown in our analysis). Consequently, we use human annotation for manual revisions. We recruit more than ten college-educated Chinese annotators. Each annotator must undergo comprehensive training to understand the annotation guidelines and meet the accuracy standards on test data before they can officially begin annotating. Given the transcription (source), model results (target), and auxiliary information, each qualified annotator must thoroughly review and annotate the entire transcript from beginning to end, to ensure consistency at the document level. They are responsible for (1) ensuring paragraph-level consistency between source and target, (2) correcting any remaining ASR errors and grammatical errors, and (3) guaranteeing that the content preserves the original content and embodies a written and formal style. During the annotation process, any confusing cases will be discussed, and the annotation guidelines and examples will be continuously updated to ensure clarity and accuracy.

Quality assurance is managed by three senior annotators. They will conduct subjective quality checks on 10% to 20% of the sampled paragraphs and utilize heuristic methods to perform rule-based inspections on all paragraphs. For documents with an error rate of less than 10%, identified and similar mistakes must be corrected. For documents with an error rate greater than 10%, a complete revision is required until the document passes the inspection.

We observe that the CoS2W task shows significant diversity. Due to labeling costs, we currently provide only one target but aim to expand target diversity in future work. Therefore, some objective metrics may not adequately reflect the performance. As a result, we have adopted metrics that focus more on semantics and introduced evaluations by LLMs and humans to mitigate the limitations of a single target.

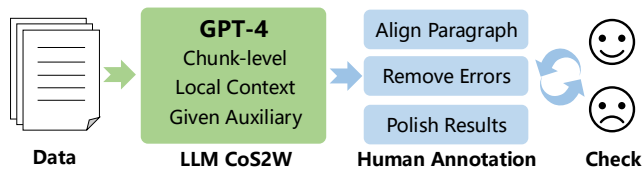


Figure 2: The pipeline of constructing the SWAB dataset.

Dataset Analysis

The target length of the final results is streamlined to 85.40% of the source length. Additionally, annotators adjust 10.43%

of the paragraph segmentation results to improve segmentation and ensure paragraph-by-paragraph consistency between the source and target. In total, 20.09% of the paragraphs are identified and corrected for ASR errors. Within the GPT-4 results, 11.29% of the paragraphs still contain ASR errors and are further modified manually by annotators. Only 2% of the paragraphs in the GPT-4 results have grammatical errors to be corrected manually. Annotators further optimize 20.26% paragraphs of GPT-4 to achieve a better written and formal style. Note that further optimization does not imply errors in the formal style of GPT-4 results.

Method

We compare the performance on the CoS2W task of different LLM across various granularity levels, and introduce contexts and auxiliary information to enhance model performance. To evaluate comprehensively, we employ automatic metrics, the LLM Evaluator, and human evaluation.

Different LLMs

To assess different LLMs’ abilities in solving this complex task, we choose 4 typical LLMs across both open-source and closed-source models. The open-source LLMs selected are QWen-14B⁶ and LLaMA3-8B⁷, chosen for their suitable computational demands for future SFT-related work. For closed-source LLMs, we pick GPT-4 (gpt-4-0125-preview)⁸ and QWen-Max (qwen-max-0107)⁹, to obtain the state-of-the-art performance of LLMs on this task.

Different Granularity Levels

To explore the impact of granularity level, we construct comparisons at the document level, chunk level, and paragraph level, ranging from coarse to fine granularity. Document-level granularity requires the model to rewrite all paragraphs of one entire document, using paragraph indexes to correlate the source and target. At the chunk-level granularity, we divide the document into chunks based on length (1.5K tokens). The model is required to rewrite each paragraph within the chunk on a paragraph-by-paragraph basis. For paragraph-level granularity, we conduct experiments focusing on the revision of individual paragraph content. Furthermore, we set chunk-level granularity as the baseline, as it allows the use of local context while requiring fewer tokens.

Context

Context is essential in text understanding, especially for spoken ASR transcripts. We divide context into local context (neighboring paragraphs) and global context (semantically related but non-adjacent paragraphs) (Han, Soldaini, and Moschitti 2021). The global context is acquired by retrieving top-k relevant paragraphs using BM25 (Robertson and Zaragoza 2009). For chunk-level experiments, paragraphs within a chunk naturally possess local context. To assess the influence of context more accurately, we conduct paragraph-level experiments. By providing 4 local or global paragraphs, we compare the effects of context.

⁶<https://www.modelscope.cn/models/qwen/Qwen-14B-Chat>

⁷<https://www.github.com/meta-llama/llama>

⁸<https://platform.openai.com/docs/models>

⁹<https://www.alibabacloud.com/help/en/model-studio/getting-started/models>

⁴<https://www.github.com/modelscope/FunASR>

⁵<https://platform.openai.com/docs/models>

Model	Auxiliary	BLEU $\uparrow$	ROUGE-L $\uparrow$	BLEURT $\uparrow$	CK-Recall $\uparrow$	S-Faithful $\uparrow$	S-Formal $\uparrow$
Source	None	16.22	41.51	55.19	12.89	7.53	4.97
LLaMA3-8B	None	8.94	20.73	40.53	13.18	6.35	7.33
QWen-14B	None	3.41	15.96	30.14	11.46	3.72	6.04
QWen-Max	None	9.47	28.92	50.58	22.35	6.00	7.55
GPT-4	None	15.51	39.66	62.15	22.49	7.32	8.20
GPT-4	+ Origin	15.33	39.53	62.73	43.70	7.38	8.09
	+ RAG	15.22	39.56	62.48	37.11	7.39	8.07
	+ Summary	15.51	39.98	62.75	40.40	7.43	8.11
	+ Keywords	15.08	39.34	62.72	45.42	7.46	8.16

Table 2: The performance comparison between closed-source LLMs and open-source LLMs at the chunk-level granularity, with different auxiliary information utilization strategies, including directly providing (Origin), retrieving the top-10 most relevant sentences (RAG), based on a summary (Summary), or a list of keywords (Keywords) derived from LLMs. Source refers to the original ASR transcripts.

Granularity	Context	BLEU $\uparrow$	ROUGE-L $\uparrow$	BLEURT $\uparrow$	CK-Recall $\uparrow$	S-Faithful $\uparrow$		S-Formal $\uparrow$	
						Para.	Chunk	Para.	Chunk
Document	None	1.80	4.97	10.64	5.30	2.28	2.44	4.57	3.74
Chunk	None	15.51	39.66	62.15	22.49	7.32	7.57	8.20	7.81
Paragraph	None	16.32	36.74	63.32	19.20	7.99	6.60	8.31	7.55
Paragraph	+ Local	16.44	39.73	67.11	22.21	8.19	7.41	8.40	7.76
	+ Global	15.55	38.73	66.00	23.64	7.89	7.05	8.36	7.74
	+ Both	16.38	39.60	67.10	23.78	8.18	7.42	8.39	7.80

Table 3: The performance of GPT-4 at different levels of granularity including document, chunk, and paragraph under different contexts. Additionally, S-Faithful and S-Formal show results for both paragraph-level and chunk-level evaluations.

We utilize auxiliary information to enhance ASR error correction by one-step and two-stage methods. The one-step method integrates full or retrieved auxiliary information as a reference part of the context in the prompt. The retrieved part is top-K relevant sentences of auxiliary information for current content based on BM25. For the two-stage approach, we first use LLMs to extract the summary or keywords from the auxiliary information, with the prompts detailed in the appendix. We require the summary to retain all key terms as much as possible, especially challenging words for ASR. The keywords should be categorized by their entity types. Next, we use either the summary or keywords as references.

Evaluations

We thoroughly evaluate using objective metrics, LLM Evaluator, and human evaluation. For objective metrics, we calculate BLEU (Papineni et al. 2002), ROUGE (Lin 2004), and BLEURT (Sellam, Das, and Parikh 2020) with the human target as reference. It’s important to emphasize that the capabilities of BLEU and ROUGE are limited by only one target of the SWAB dataset. Therefore, we focus more on semantic-oriented automatic metrics such as BLEURT.

To measure the performance of ASR Error Correction, we design the Challenging Keyword Recall (CK-Recall) metric. The challenging keywords are a subset of ASR errors in transcripts, focusing specifically on named entities (e.g., person names, podcast titles). Correcting these errors can significantly improve the reading experience. Furthermore, these words typically cannot be modified or adjusted, making it convenient to directly search in the results. We use the recall calculation formula, where the numerator represents the

number of keywords that appear in the results, and the denominator represents the total number of keywords.

In addition, we utilize an LLM (i.e., GPT-4) as the evaluator with the prompts shown in the appendix. We request the LLM to evaluate and score the model-generated target across faithfulness (S-Faithful) and formality (S-Formal). Scores range from 1 (worst) to 10 (best) for each criterion. S-Faithful reflects content preservation to retain original meaning, and S-Formal reflects the degree of formality in style. For chunk-level results, we require the outputs to align each paragraph by the paragraph index, so we evaluate them at the paragraph-level granularity. To fairly compare paragraph-level and chunk-level results, we evaluate them both at paragraph and chunk levels. To enhance effectiveness, we sample a maximum of 100 paragraphs per document for evaluation. Furthermore, we sample over 200 paragraphs, whose results of different LLMs are ranked by human evaluators.

Results

Results of Different LLMs

As shown in Table 2, we compare GPT-4 and QWen-Max (closed-source LLMs) with LLaMA3-8B and QWen-14B (open-source LLMs) at the chunk-level granularity. GPT-4 outperforms all other models across all metrics, followed by QWen-Max. Due to the limitations in model size, the performance of open-source models isn’t as good. Among them, the LLaMA3-8B model exhibits superior performance compared to the QWen-14B model.

LLMs show potential in solving this complex task, yet still encounter “faithfulness problems”. The performance of LLMs on the faithfulness score is unsatisfactory, even for

state-of-the-art models like GPT-4 (7.32). We observe some “paragraph drift” situations at the chunk level. For example, the end of one paragraph might be migrated to the beginning of the following one. Even worse, the target and source cannot be aligned by paragraph index. Furthermore, faithfulness problems occur even when alignments are correct. The model frequently prioritizes coherence and fluency over faithfulness, such as fabrications for fragments or the omission of disorganized insertions in spoken scenarios, which compromises the accuracy of some essential information.

Results of Granularity Levels and Contexts

As presented in Table 3, we compare the performance at different levels of granularity based on the GPT-4 model. Although the SWAB dataset is constructed at the document level, the CoS2W task at the **document level still presents a challenge**. It requires a considerable number of tokens for both input and output (about 8K), which is difficult even for GPT-4 as shown in Table 4. For document-level results, 76.10% of the paragraphs are blank without the corresponding index, compared to 1.60% at the chunk-level granularity. And there is a noticeable decline in all metrics.

Due to “paragraph drift” issues at the chunk level as mentioned before, we present LLM evaluation results (i.e., S-Faithful and S-Formal) for both paragraph and chunk levels to ensure a fair comparison. The chunk-level evaluations show that the CoS2W results perform better at chunk-level (7.57 and 7.81) than at paragraph-level (6.60 and 7.55) granularity. This highlights the **advantages of utilizing local context at the chunk level**. Chunk-level granularity enables LLMs to naturally leverage the local contexts to enhance faithfulness and formality performance.

In addition, we experiment with various contexts at paragraph level explicitly, **confirming the role of context in enhancing text understanding**, as is shown in Table 3. Paragraph-level results with local context show improvements across all metrics compared to results without context, approaching chunk-level performance. This further verifies the supportive role of local context. Local context improves CK-Recall from 19.20% to 22.21%, assisting the model to better understand the current paragraph. Moreover, global context can also yield some benefits, while not as significant as local context. Combining both local and global contexts, further improvements are seen in some metrics.

Results of Auxiliary Information

We employ multiple methods to leverage auxiliary information. Table 2 includes results of directly providing (Origin), retrieving the top-10 most relevant sentences (RAG), based on a summary (Summary), or list of keywords (Keywords) derived from LLMs.

Auxiliary information can enhance ASR error correction, while requiring more effective methods. Based on the GPT-4 model at chunk-level granularity, directly given auxiliary information (Origin) increases CK-Recall from 22.49% to 43.70%. The improvement of RAG is limited, indicating a need for better retrieval methods. The Summary method improves objective metrics but shows limited CK-Recall gains, possibly due to missing keywords in the summary. The Keywords method is most effective, further

Model	#Tokens	BLEURT↑	Blank%↓
GPT-4 (paragraph)	340.50	63.32	0.00
w/ local context	654.47	67.11	0.00
GPT-4 (chunk)	1515.00	62.15	1.60
w/ auxiliary	2006.86	62.73	1.28
GPT-4 (document)	8140.42	10.64	76.10
QWen-14B (paragraph)	340.50	59.87	0.00
w/ local context	654.47	55.64	0.00
QWen-14B (chunk)	1515.00	30.14	36.56
w/ auxiliary	2006.86	22.35	48.33
QWen-14B (document)	8140.42	2.87	91.58

Table 4: The comparison of instruction-following abilities between GPT-4 and QWen-14B. Instructions are arranged by increasing difficulty levels and the number of input tokens. The fewer the blank paragraphs, the stronger the instruction-following ability.

Model	S-Faithful↑		S-Formal↑	
	Total	Non-Blank	Total	Non-Blank
GPT-4	7.32	7.39	8.20	8.29
QWen-Max	6.00	6.50	7.55	8.21
QWen-14B	3.72	4.99	6.04	8.29
LLaMA-8B	6.35	6.53	7.33	7.56

Table 5: Comparison of different LLMs among total paragraphs with non-blank paragraphs at the chunk-level granularity.

boosting recall to 45.42%, and enhancing faithfulness scores while maintaining formality scores basically. However, improvements in CK-Recall stay below expectations, suggesting the limited benefits of current methods. Effectively utilizing auxiliary information still needs further research.

Analysis

Instruction Following Capability

We observe that **larger LLMs (e.g., GPT-4) show stronger instruction-following ability** compared to smaller LLMs (e.g., QWen-14B). We instruct the models to organize their responses by aligning each paragraph with the corresponding content using the paragraph index. We refer to paragraphs without aligned paragraph indices as *blank paragraphs*, which do not have corresponding target results. The rate of blank paragraphs can reflect a model’s instruction-following capability. The fewer the blank paragraphs, the stronger the instruction-following ability. Thus we compute the rate of blank paragraphs as shown in Table 4. At the chunk-level granularity, GPT-4 demonstrates robust alignment with scarcely any blank paragraphs (1.60%), whereas QWen-14B exhibits more alignment errors (36.56%). Given the auxiliary information, as the number of input tokens increases, QWen-14B’s performance further declines to 48.33%, while GPT-4 remains virtually unaffected.

The differences in instruction-following ability also impact the model’s performance on the CoS2W task. As shown in Table 5, we compare the results among total paragraphs with non-blank paragraphs. Among non-blank paragraphs, QWen-14B’s S-Formal performance is actually quite good (8.29). However, the overall performance (6.04) is relatively low due to the influence of blank paragraphs.